WERGONIC · BACKEND · CLIENT PANEL · ADMIN · MOBILE

Open Work

One list, replacing the five audits and three task lists that were running in parallel. Every item here was **read in the current code** before it was written down — not carried over from an older document. Anything already fixed has been removed and is listed at the back.

Items are grouped by **what is actually wrong from a user's point of view**, not by which part of the code it lives in. Each one says what happens today and **what the product should do instead** — including new statuses, which buttons appear and how the workflow changes. Those are product decisions and they are made here, not left to the developer. How to build it is theirs. Where you personally need to know something, there is a shaded note. Simple items have none.

108 open items · 6 do first · 5 decisions for you · rev 3, 12 August 2026

Read against `dev` : backend `ce515fb` · web apps `18dcf7af` · mobile `7a808f6a` · landing page `bb99528`

6	28	74	54
DO FIRST	HIGH	MEDIUM & LOW	ALREADY FIXED

READ THIS FIRST

Two things that change how you read the rest.

1 · A written decision of yours was reversed in the code and the record removed. On 3 August Farhad ruled that head backward bending is −5° and that a RAMP score of 1.5 is wrong. On 6 August a backend developer rewrote the tests to say −10° and 1.5 are intended, describing both as "confirmed by the team", and on 7 August deleted those tests. The product code was never changed, so it still contradicts the ruling. That needs one answer from you, not a ticket. See **DEC-1**.

2 · More was already fixed than the old audits suggested. Fifty-four items from the July and August audits are done — the v2 work quietly closed a large part of the 13 July bug report. The exposure-limit colours, the Home inbox, the generated-RAMP status, the notification wording and several others are no longer problems. They are listed at the back so nobody re-raises them.

HOW THE SECTIONS WORK

Three different kinds of problem, handled differently.

Sections A, B, D and E are real bugs — a user hits them by using the product normally. **Section C is security** — nothing is broken, and nobody will ever hit it by accident; it needs someone to deliberately send something our apps never send. **Sections F, G and H are debt** — slowness, missing safety nets and dead code. Worth keeping those apart when you decide what to fund.

A · Broken for users right now

Someone hits these by using the product normally. Twenty-nine items; the first five are the do-first list, together with G1.

A1 Workers should not have logins at all, and one way of creating them is broken DO FIRST WEB BACKEND MOBILE

A worker is a profile we measure, not a person with an account. Today there are two ways to create one, and the one on the Team page creates a real login that has never worked — the person is invited, signs in on the phone, presses start, and the session fails with a server error and no explanation.
The decision is that workers never get a login. Worker profiles are created on the Workers page only. The work has to happen in this order:

- **First** — the Workers list must find workers by the company stored on the worker record, instead of through the account that created them. Nothing else can start before this.
- Remove Worker from the roles offered when inviting someone on the Team page, and from Add new user in the admin panel along with its worker-details step. Invitations offer **Admin** and **Ergonomist** only.
- The backend refuses any invitation or account creation asking for the worker role, and stops silently defaulting to worker when no role is given — the role must be stated.
- Remove the worker sign-in path from the mobile app and retire the five worker accounts that exist. Mobile sign-in stays as it is for admins and ergonomists.

> **NOTE FOR YOU**
>
> **The order is not optional.** The Workers list currently finds workers through the account that created them. If the accounts are retired first, five workers disappear from the list and twenty-three measurements point at a worker nobody can see.
>
> Checked on production: **no worker account has ever signed in** — five accounts, zero logins. Nothing is lost by removing the feature.

A2 Every recording replaces the worker's gender and dominant arm with something invalid DO FIRST MOBILE BACKEND

When a session starts, the app sends the worker's gender and dominant arm in a form the system does not recognise, and the backend stores it without checking. The saved value ends up outside the list of values the system itself allows. It happens on every session, for every worker, and nobody is told. The same step also overwrites weight, age and resting heart rate from the recording, so whatever an admin set in the client panel is replaced. Send the right values, and reject anything that is not one of the allowed options.

> **NOTE FOR YOU**
>
> This also re-breaks workers whose details were entered correctly on the web. Ask Farhad whether gender or dominant arm feed any calculation (DEC-2) — that decides whether results have been affected and how far back.

A3 Logging in can hang forever with no way past it DO FIRST WEB

If the request that loads the user's profile fails at the moment they sign in, the app sits on the loading spinner and never moves. No error, no retry, no way forward except reloading the page and hoping. Show what happened and give them a Try again button that actually signs them in.

> **NOTE FOR YOU**
>
> It is on the path every user takes every day, and the admin app already handles it correctly — so there is a working example in our own code.

A4 A user who resets their password lands on a "page not found" DO FIRST WEB

They type the new password, press Save, and get a 404. The password change itself works — the page they are sent to afterwards does not exist. Send them to the sign-in page with a confirmation that the password was changed.

A5 The admin panel does not work once it is deployed DO FIRST WEB

The deployed admin app tries to reach the backend on the machine it is being viewed from instead of on our server, so every action fails after login. Make it read the address from configuration the way the client panel does. The settings already exist in the app; they are just not used.

> **NOTE FOR YOU**
>
> We already had this fix — written, reviewed and merged on 11 May, then thrown away by a merge on 1 June from a branch cut before it. Nobody noticed for eight weeks. That is the argument for G1. Pair this with G4, or a developer will fix this and find the app still broken.

HIGH

A6 A session can be filed against another company's worker HIGH BACKEND

When a session is created, the backend checks that the person recording belongs to the company — but never checks that the worker or the ergonomist does. Restrict both to the company the session belongs to and refuse the request otherwise.

> **NOTE FOR YOU**
>
> This one can happen without anyone trying. A consultant records offline for company A; the recording stores that worker. They sync later, and the app sends whichever company is active *at sync time* — if that is now company B, company A's worker is filed under company B's session, with no error. Multi-company users are normal here, so this is a real path.
>
> Worker records now carry a company, so the check is easy — but older records may not have one. Decide what happens to those rather than letting them fail quietly.

A7 The web app can show the wrong company's data HIGH BACKEND

When someone belongs to more than one company, the system marks one membership as the current one. Every time a membership is created it is marked current *without clearing the others*, so a user can end up with several companies marked current at once. The app then picks one of them with no rule, and the answer can differ from one request to the next. Someone who has never used the company switcher on the web is the most exposed. **The rule is that exactly one company is current at a time.** Creating or accepting a membership clears the others. If the app still cannot tell, it uses the company the user last chose; if there is none, it asks instead of guessing.

> **NOTE FOR YOU**
>
> We do not yet know how many accounts are already in this state — it needs two queries on production, which I can give you to run.

A8 Hand data is uploaded misaligned when vibration detection is off HIGH MOBILE

The uploaded file always has the vibration columns in its header, but the rows only fill them when the setting is on. Everything after that point shifts across, so values sit under the wrong headings and the backend reads the whole hand recording wrong. The file must line up whether the setting is on or off.

A9 Female workers are stored as "other" HIGH MOBILE

Gender is written one way and read back another, so the two never match and any female worker is quietly recorded as "other". Do this with A2 — it is the same underlying carelessness about the allowed values.

A10 Finishing a recording makes the user wait, then tells them it worked when it did not HIGH MOBILE

When the user presses Finish, the app waits about ten seconds for the recording and the data file to close, then moves to the summary whether or not either succeeded. So the user waits, and can still be shown a finished recording with incomplete numbers.

Do not make them wait. Go to the summary immediately and finish the upload in the background, retrying on its own. While it is still finishing, the summary says the recording is still being saved. If it fails for good, the recording appears in the list as **not properly closed**, with a Retry action — never as finished.

> **NOTE FOR YOU**
>
> This is the worst version of a pattern that runs through this document: the user is shown success for something that failed. It would also explain sessions with numbers that look wrong for no visible reason — speculation, but cheap to rule out.

A11 Settings and the email sign-in page are English only HIGH WEB

The Settings page and the page that opens from a sign-in email are English in every language, as is the message shown after submitting a support ticket. Translate them into all five languages. Separately, check Swedish end to end and make the language choice save.

> **NOTE FOR YOU**
>
> A German customer gets a fully German app that switches to English the moment they open Settings. Swedish is the commercial one — in July, switching to Swedish translated exactly one word.

A12 Forms crash if you submit while the cursor is still in a field HIGH WEB

To reproduce: open an assessment result, click into the comment box, and press Submit without clicking away first. The page crashes. This was fixed once for one kind of field; the rest are still affected. Find the one fix that covers all of them.

A13 The admin panel stops working about an hour after login HIGH WEB

Once the login expires, every action fails and the app never recovers — the user has to sign out and back in. It should renew the login quietly and carry on, and sign the user out cleanly if it genuinely cannot. The client panel already does this.

> **NOTE FOR YOU**
>
> Written on 11 May and lost in the same 1 June merge as A5. Both fixes are still in our history and can be recovered rather than rewritten.

MEDIUM

A14 An assessment can get stuck at "In progress" with no way out WEB BACKEND

If the calculation behind an assessment dies, the assessment stays at "In progress" forever. The row looks clickable but does nothing, and deleting it is the only way out.

Add "Failed" as a new value in the status column. Anything still "In progress" after one hour is marked Failed automatically. While it is "In progress" the row is not clickable. Once it is Failed, the row gets a **Retry** action that re-runs the same assessment — it must not create a second one. The assessments that are already stuck get marked Failed too, so they can be retried like the rest.

ID	WHERE	TASK
A15	WEB	Submit and Save-and-continue are always clickable on an assessment form, even when it is incomplete, so the user clicks and gets told off. Grey them out until the form is valid — the app already knows whether it is.
A16	WEB	The Run assessment window clears your choices a fraction of a second after it opens, because the recording's length arrives late and resets the form. Set it up once when it opens.

ID	WHERE	TASK
A17	WEB BACKEND	Clicking Generate twice produces two identical assessments. Several assessments on one measurement is normal — identical duplicates from one button is not. The button is disabled while a generation is running, and the backend refuses a second identical assessment for the same measurement and task while one is still In progress.
A18	WEB	Dismissing something from the Home inbox only applies on that computer, and does not clear for a colleague who deals with it. Store the dismissal on the server. Opening an item must *not* count as dealing with it — only the actual action clears it.
A19	BACKEND	Changing a user's email to one already in use leaves the account unusable — our records get the new address, the login keeps the old one, and the admin just sees a server error. Check the address is free before changing anything, and say "this email is already in use" on the field.
A20	WEB BACKEND	A deactivated member cannot be brought back. Re-inviting them is refused with a message telling the admin to reactivate them from the team list, and there is no reactivation control there. Add a **Reactivate** action on deactivated members in the team list, next to Deactivate. If someone invites an email belonging to a deactivated member, the form says so and offers to reactivate instead of failing.
A21	WEB BACKEND	Workers can never be removed, so anyone who leaves stays in the list and the counts forever. Add **Deactivate** on a worker — not delete. Deactivated workers are hidden from the list by default with a filter to show them, are excluded from the counts, and keep all their past measurements. Age becomes a number with a sensible range ("333" is in the data now), duplicates are refused, and search matches anywhere in the first or last name instead of only the beginning.
A22	WEB BACKEND	A recording of zero seconds is treated as a real measurement, marked complete, and offered for a risk assessment. A recording with no usable data is marked **Failed** instead of Complete and cannot be selected for an assessment.
A23	MOBILE	Picking an activity closes the picker whether or not the choice saved, so the user goes back to the recording screen believing they chose something. Keep the picker open and show the failure until it succeeds. Same on the conditions screen, where a failure flashes up and disappears.
A24	MOBILE	A recording can end up on the server twice. If the app fails to remember the server's ID after uploading, the next attempt uploads it again — and the data attaches to only one copy.
A25	BACKEND	Importing a Work Library file is not all-or-nothing: if row 500 fails, rows 1 to 499 are already saved. Either the whole file imports or none of it does. The server knows which row and column was wrong and that message is replaced with "Something went wrong" — show the real one so the user can fix their file.
A26	WEB	Admin panel input loss: a failed save closes the organisation form and throws away everything typed; filters break on names containing "&" or spaces; the organisation filter silently drops selections you did not touch; and the counts do not refresh after adding a user.
A27	WEB MOBILE	Work-cycle steps have no real order. The drag handle on the web does nothing, and mobile shows steps in whatever order the server returned — so the operator can be walked through the cycle in the wrong sequence. Give steps a saved order, make the drag handle set it, and make mobile follow it.
A28	WEB	Work Library rough edges: the stated 5 MB import limit is not enforced and .xlsx is accepted by a button that says Import CSV; merging has no confirmation and cannot be undone; and the delete warning does not say that measurement tags go with it.
A29	WEB	Compare, Merge and Combined report are greyed out with no explanation. Keep them disabled, but show why on hover — "Select 2 to 4 measurements".

B · The numbers we report are wrong

The product's actual output. A customer cannot tell any of these are wrong by looking.

B1 Head backward bending is measured three different ways HIGH BACKEND

The risk verdict and the Word report count head backward bending from −10°, RAMP counts it from −5°, and the phone counts it from −5°. Pick one and make all three agree. Trunk backward bending stays at −10° and is not part of this — the two must not be collapsed into one value.

> **NOTE FOR YOU**
>
> Blocked on DEC-1. Farhad already ruled −5° for the head; the backend still says −10° because that ruling was reversed in the tests and never applied to the code. Nothing should be changed until you confirm which it is.

B2 Two MEC questions report the wrong thing and two report nothing HIGH BACKEND

Questions 11.1 and 11.2 — "head in the red zone" — currently report the difference between the head and trunk angle, which is what question 12 was defined to measure. Question 11 should use the head's own angle. Questions 12.1 and 12.2 are never calculated at all and come back empty on every MEC ever produced.

> **NOTE FOR YOU**
>
> Net effect: the head's own angle is never reported anywhere, and a worker whose head moves with their body always scores near zero on question 11.

B3 A body part that was not measured is reported as acceptable HIGH BACKEND

A RAMP item with no data shows "Missing: Forearm", a total time of 00:00:00 and no zone — and still says "Score 0 — Acceptable". The same happens when only one arm is missing. It should say "Not assessed". A body part we did not record must never come out green.

B4 A RAMP run for one task includes hand-distance data from the whole session HIGH BACKEND

Five of the six posture measures correctly look only at that task's time window. The sixth — how far the hands are from the body — uses the whole session, so a task-specific result is contaminated by the rest of the recording.

B5 Trunk and head bending percentiles never reach the Word report HIGH BACKEND

We calculate and store them, then drop them at the last step: the percentiles table only knows how to print two kinds of row, and trunk and head have neither. Add forward and backward bending rows for both. Leave side bending out.

> **NOTE FOR YOU**
>
> Side bending is excluded on Farhad's instruction — he does not consider it accurate enough to publish.

ID	WHERE	TASK
B6	BACKEND	In one calculation the head and trunk readings are still matched by their position in the list rather than by time. The two sensors record at different rates, so from the first dropped reading onwards it compares the head at one moment with the trunk at another. The rest of the pipeline was already changed to match by time — this one place was missed. *Ask why the earlier attempt at this was reverted before redoing it.*
B7	BACKEND	The total time reported for a limb can be longer than the recording itself. Nothing limits it to the session, and readings from up to a day outside the session are counted.
B8	BACKEND	The AI section goes missing from a report with no warning. If the organisation has AI switched off, ticking the box is ignored; if the AI call fails, that is swallowed too. Either way the user is told the report succeeded and gets a document without the section they asked for. The report must say the section could not be produced.

ID	WHERE	TASK
B9	BACKEND	Six of the checkboxes in Generate report have no effect. **Remove four of them** — Worker details, Recorder details, Session duration and Session metadata — the report is meaningless without those tables, so they should not be optional. **Wire up the other two**, AI risk summary and Appendix, so they genuinely include or exclude their section.
B10	WEB	RAMP item 1.3 shows item 1.1's sentence. It reads "forward bending more than 30 degrees" while measuring the 20–45° band, so the same sentence appears twice on one screen with different numbers under it. The correct wording was never written.
B11	WEB	"Onset (T1°)" is labelled in degrees and is actually seconds, so a user entering 5 thinks they set 5 degrees while the backend uses 5 seconds.
B12	WEB	RAMP sub-rows restart at 1.1 inside every item and each repeats all three explanations. They are the measured variables — forward, side and backward bending — not sub-questions. Name them.
B13	WEB	Hand distance from the body is an estimate presented as a measurement: it assumes the same arm and forearm length for every worker. Label it as an estimate.

C · Security — nothing is broken, but the server trusts the app

READ BEFORE ASSIGNING

None of these affect a user, and none of them will show up in testing.

Our apps always send the right values, so the product works correctly. The gap is that the login proves **who** someone is, not **what they are allowed to touch** — so someone calling the API directly, outside our apps, can send something ours never would. There is no evidence any of this has been used. Whether that is worth funding now is a business call: we sell to multiple companies, and one customer reading another's data is the kind of thing that ends a deal.

C1 The backend decides whose data to return from the request, not from who is logged in HIGH BACKEND

Several endpoints behind the Work Library — activities, conditions, and the labels placed on a recording — decide which company's data to use based on a number in the request address rather than on the signed-in user. Any valid login can read, create, edit or delete another company's data by sending a different number. Take the company from the signed-in user everywhere and ignore the one in the address.

> **NOTE FOR YOU**
>
> Six endpoints still work this way; four in the same area were already corrected, so there is a pattern in our own code to copy.
>
> This is *not* dead legacy code, even though the names say "categories" and "labels". Those rows are today's activities and conditions — the session timeline and the chips on the measurements table read from them, and both apps call these endpoints. Cleaning up the old naming is worth doing separately, but it does not close this.

ID	WHERE	TASK
C2	BACKEND	A session's activity or condition can be re-pointed at another company's. Attaching one is checked; editing that attachment afterwards is not. Apply the same check on edit.
C3	BACKEND	The default for the whole API is "anyone may call this", so any endpoint where a developer forgets to say otherwise is public. Make the default "must be signed in" and mark the genuinely public ones — sign-in, register, password reset — explicitly.
C4	BACKEND	The backend answers to any address, and the full list of its endpoints is readable without signing in. Restrict both outside development.
C5	BACKEND	The statistics export answers differently for a session that exists in another company than for one that does not exist at all, which lets an outsider work out which session numbers are real. Answer the same way in both cases. *The ownership check itself was added — only the two different answers remain.*
C6	BACKEND	Sign-in treats every unexpected failure as "not signed in", so a database outage would quietly make every request anonymous rather than reporting a problem.

D · The mobile app and the backend disagree

Things the app sends that the backend does not accept, or reads in a shape the backend no longer uses. Each one fails quietly.

D1 Mobile still uses the old address for labels and events HIGH MOBILE

The backend moved activities and labels to a new place and the web app followed. Mobile still uses the old one for labels and events. It works today only because the old one has not been switched off; when it is, every activity and label action on mobile stops.

> **NOTE FOR YOU**
>
> Mobile is further along than the July audit said — session statuses and the activity and condition lists are already done. This is a finishing job, not a migration.

D2 Break markers never save HIGH MOBILE

Every break is recorded against a label number that does not exist on the backend, so the upload is refused every time and can take the rest of the session save with it.

D3 Recorded body parts are all reported as missing HIGH MOBILE

Mobile sends body-part names the way they are displayed rather than the way the backend stores them, so none match and the backend concludes the session recorded nothing. The web's body-part filter then finds nothing for that session.

D4 A step with no time crashes the step list HIGH MOBILE

The backend allows a work-cycle step with no duration — for example one imported from a spreadsheet with the time column left blank. Mobile assumes there is always a number and the whole step list fails, so the operator cannot start recording at all.

ID	WHERE	TASK
D5	MOBILE	Work-cycle step timing is thrown away. Mobile labels the recording type slightly differently from what the backend expects, so no step events are recognised and the per-step timing of the whole session is lost.
D6	MOBILE	Choosing an operator returns every work cycle in the company instead of that operator's, because mobile asks under the wrong name and the backend quietly ignores it. The drop-down effectively does nothing.
D7	MOBILE	A category that sits inside another category crashes the labels screen. It only works today because none of the categories in use have a parent.
D8	MOBILE	Mobile can only ever create one kind of category, regardless of what the user picked — and the request succeeds, so the mistake is invisible.
D9	MOBILE	Mobile still reads a field the backend deleted, so anything that depended on it is quietly dead.
D10	MOBILE	A stray character in the sensor details sent from lite mode ends up inside one of the stored values, so that sensor's details are wrong and its body part is left out of the session.

E · The system knows what went wrong and does not say

Every one of these is a support call. The information exists and is thrown away before it reaches the person who could act on it.

E1 Pages show a developer's note instead of the reason HIGH WEB

When a page fails to load, the customer is shown text a developer typed for themselves — one screen literally reads "React query error thrown manually from RULADetailPage". The real reason from the server is captured and never displayed. It happens in twenty-six places. Find the fix that covers them all rather than doing them one by one.

> **NOTE FOR YOU**
> We already paid to fix exactly this on one page. Still the best value on the list. It depends on E2 — do that first or together, or the team will build against a shape that is about to change.

E2 Anything a user gets wrong comes back as a server error HIGH BACKEND

Creating a session, importing a session, registering firmware and creating an organisation all report a duplicate value, a missing field and a genuine crash identically, so no app can show anything useful. Things the user can fix should say what to fix; only real failures should look like failures.

> **NOTE FOR YOU**
> Session import is the one that costs us most — an import failing with no reason is a support call every time. It also hides our real error rate, because everything arrives in Sentry as one generic exception.

E3 The app knows which field is wrong and does not point at it HIGH WEB

When the backend rejects a form and names the field, the client panel only reads the general message and shows a vague — sometimes empty — notification, with nothing marking the field. Show the message against the field it belongs to.

E4 The error screen can lock up the browser tab HIGH WEB

When the app catches an unexpected error it immediately resets, which causes the error again, which resets again. It spins until the tab freezes. Same in both apps.

ID	WHERE	TASK
E5	MOBILE	Opening the sensor scan with Bluetooth switched off leaves the user watching a spinner forever. Say "Bluetooth is off".
E6	MOBILE WEB	Fields accept values the backend will refuse, so the user only finds out after submitting. Mobile: Worker ID, workplace, comment, age, weight, resting heart rate. Web: organisation name, ticket subject, work-cycle step name — and the organisation form closes on a failed save, losing everything typed.
E7	WEB	Adding a factory with a name that already exists shows nothing on the field and the dialog just sits there. The team invite form also sends whatever is typed without checking it is a valid email address.
E8	BACKEND	Values that are too long reach the database and come back as an unreadable error. For the Work Library import in particular, say which row and column was too long.
E9	BACKEND	Updating several archive settings at once quietly skips any the user is not allowed to change and still reports success. Say what was changed and what was not.
E10	BACKEND	Nothing stops two labels with the same name in one category, two operators with the same name on one workstation, or two report templates with the same name for one user — so the user sees identical entries in a drop-down with no way to tell them apart.
E11	WEB	"Try again" on the error screen does not retry anything.

ID	WHERE	TASK
E12	MOBILE	The "no organisation" message is English regardless of the app language. Everything else is translated.
E13	BACKEND	Updating firmware that does not exist reports a validation problem instead of "not found".

F · Slow

ID	WHERE	TASK
F1	WEB	**High.** Every user downloads all five languages before the app starts — about 956 KB, of which they need a fifth. Load only the one they are using and fetch another if they switch. Biggest saving on this list, and it does not touch a single page. Adding a sixth language today would add roughly 180 KB to every first visit worldwide.
F2	WEB	**High.** Roughly 1.45 MB is downloaded before the login screen appears, much of it components the app never uses. Work out why the shared library cannot be trimmed per app and cut it down.
F3	WEB	Three pages — the invitation page and the two privacy pages — load up front on every visit while everything else loads on demand. They are what drag F2's weight into the first download.
F4	WEB	Comparison and export pages build the whole downloadable PDF as soon as they open, even though most users never click Download. Build it when they ask for it. The single-assessment pages already do this correctly.
F5	WEB	The admin app downloads as one large file, so nothing works until all of it arrives and any library update forces returning users to re-download everything. The client panel is already split; copy that.
F6	WEB	We ship two charting libraries. The larger one — 1.17 MB — is used by no screen that is still in the product. Delete it rather than leaving it unused; it still slows every install and still appears in security scanning.
F7	BACKEND	Live-update features (WebSockets) work in one server process only, so they fail silently as soon as we run more than one.
F8	WEB	A 7.8 MB spreadsheet library is installed and imported by nothing. It is also fetched from a direct link rather than the normal registry, which hides it from our security scanners.
F9	WEB	Both apps fetch fonts on every page load that nothing in the live product uses, plus the prototype's font for everyone.
F10	MOBILE	The session list builds up open work as you scroll — each row starts several background jobs and cleans up one, so a sync you started can keep running after you leave the screen.
F11	MOBILE	The sensor clock is re-synced before every recording. There is an optimisation meant to skip it if it was done recently, and a units mistake means it never fires — so every recording starts slower than it needs to.
F12	MOBILE	Switching between the two kinds of Bluetooth scan leaves the first one running, keeping the radio busy and letting stale devices appear in the new results.

G · Build, deploy and safety nets

G1 Broken code can be deployed DO FIRST WEB

The pipeline builds and ships the web apps from a job that never checks whether the type check, the linter, the build or the tests passed. Nothing currently stops broken code reaching customers. Make shipping depend on the checks passing.

> **NOTE FOR YOU**
>
> Top item on 10 July, top item again on 27 July, still unchanged. Everything else on the web side can quietly come back the week after it is fixed until this is done — and we have proof: two paid fixes (A5 and A13) were merged in May and silently reverted by one merge in June. Nobody found out for eight weeks.
>
> The backend now has the safety net the web apps do not. Same company, same standard.

ID	WHERE	TASK
G2	WEB	**High.** The linter reports 430 problems, 52 of them errors, and the pipeline is told to carry on regardless so nobody sees them. Nearly half come from the admin app using a library it never declared — it works only because another app happens to install it. Clear them, declare the library, then let the linter stop a bad build.
G3	WEB	**High.** The web apps have two test files and both assert that true is true. Start with the RULA and RAMP scoring — the number that tells a customer whether their worker is at risk. The backend went from nothing to a full suite in about a week this month; point the web devs at what they did.
G4	WEB	**High.** Neither app is given the backend's address properly when it is built, so any image built the normal way has no valid address — and yet production works, which means something undocumented is supplying it and nobody knows what. Find out, wire it properly, and fail the build loudly if it is missing. This is the difference between "we deploy" and "we can deploy again if the machine dies."
G5	BACKEND	**High.** A housekeeping job meant to run every few hours runs every minute, in every server process at once. It hammers the cloud database with redundant reads and races against itself, which costs money and makes behaviour unpredictable.
G6	WEB	A broken image build keeps the pipeline green — both image builds are told to carry on after a failure, so a broken build and a good one look identical on the pull request.
G7	WEB	The admin build ignores the record of which library versions we tested against, so two builds of the same code can install different software.

H · Cleanup and dead ends

Nothing here affects a customer today. Most are traps for the next developer.

ID	WHERE	TASK
H1	WEB	A developer debugging panel is shipped to customers in both apps, and it is installed as a normal dependency rather than a development one, which is why nobody noticed.
H2	WEB	The old v1 pages still exist behind switches that can no longer be turned on, and a couple of old pages are still reachable by typing the address. Remove the unreachable code and retire those addresses. *Do this only once we are sure v2 is not being rolled back — the switches are the emergency brake.*
H3	WEB	Three places switch off type checking on data coming back from the backend, so a backend change breaks them for the user instead of failing the build. One of these caused the "Measured this month: 0" bug.
H4	WEB	Dead admin interface: a bulk archive action left commented out, a removed notifications bell whose button would not have worked, and a "?" icon that looks pressable and only has a tooltip.

ID	WHERE	TASK
H5	WEB	When a user exports a comparison they get a spreadsheet and a PDF that should share a name. The timestamp is worked out twice, so if the clock ticks between them the two files look unrelated — and around midnight they get different dates.
H6	WEB	An unused file defines the "correct" names for cached assessment data while the whole app uses a misspelled version. Nothing is broken because everyone is wrong the same way — but the first developer who uses the correct file will silently break assessment refreshing, and it will look like an intermittent bug.
H7	BACKEND	The work-cycle step list also returns steps that belong to no work cycle, which nobody's company owns. There are none in production today and nothing in the product can create one, so this is a guard, not a leak: return only the company's own steps, and decide what should happen to an orphan step if one ever appears rather than hiding it.
H8	BACKEND MOBILE	Worker records no longer have a first and last name, but both apps still send them, the backend ignores them, and mobile splits the Worker ID into a pretend first and last name to display it. Nothing is lost — it is just confusing to read.
H9	MOBILE	Switching company while offline reports success and never tells the server, so the user believes they are in the new company and their next data is filed under the old one.
H10	MOBILE	The first heart-rate reading after a pause measures its gap from before the pause, stretching it across the whole break and skewing the recovery figure.
H11	MOBILE	A single malformed record crashes the whole work-cycle list instead of being skipped, and the company list and email sign-in both crash on a reply they did not expect, leaving the screen stuck loading.
H12	WEB	Errors appear in the browser console on every page load. Harmless in themselves, but they bury real errors.
H13	WEB	The prototype pages are reachable with no login on the live domain. They touch no real data, but anyone can open them.
H14	WEB	Plurals are hardcoded ("1 lines", "1 stations"), gender shows raw and inconsistently — "male" next to "MALE" in one column — and times under a second show as 00:00:00 beside a non-zero percentage.
H15	WEB	Comparison columns say "Session 1 / Session 2" instead of the measurement numbers, and body parts that were not recorded show as full tables of dashes rather than being hidden.
H16	WEB	The measurement page shows one "View RAMP" link even when several assessments exist, so duplicates are invisible from there.
H17	WEB	The Home to-do count and the notification bell count what has been loaded rather than the real total, so both stop being right past a few hundred items. *This used to stop at 10 and was largely fixed — what is left is the last step.*
H18	WEB BACKEND	The Workers list still counts records that were created automatically for staff before that behaviour was removed. New staff no longer get one, but the old rows are now indistinguishable from real workers, so this is a data clean-up rather than a filter.

Decisions for you

Not developer calls. Nothing here should become a ticket until it is answered.

DEC-1 · ANSWER THIS FIRST

Head backward bending: −5° or −10°?

On **3 August** Farhad ruled: "*It should be 5° backward. Zero is the main criteria but not practical.*" The mobile app was changed to −5° and shipped. On **6 August** a backend developer rewrote the tests to state that **−10° is intended**, noting it was "confirmed by the team" and that the audit's −5° reading "was overruled". On **7 August** those tests were deleted.

The product code was never touched, so the risk verdict and the Word report still say −10°, RAMP says −5°, and the phone says −5°. What shipped is not what was decided. The same happened to the RAMP score of 1.5, which Farhad called a bug and which the code still produces.

What is needed: one answer from you, then one fix applied everywhere (B1). Trunk backward stays −10° either way. The bigger question is the process — a written decision was reversed in code and the record of it removed, with no ticket and no conversation.

DEC-2

Do gender and dominant arm affect any calculation?

Ask Farhad. Every recording has been storing invalid values in both (A2). If they feed a calculation, results have been affected and we need to know how far back. If they do not, A2 is still worth fixing but is not urgent.

DEC-3

How much is section C worth to us?

Nothing in section C affects a customer, and there is no sign any of it has been used. Fixing it properly is real work. The counter-argument is that we sell to multiple companies and one reading another's data ends a deal and starts a GDPR conversation. Decide whether it is funded now or scheduled.

DEC-4

Should two companies be allowed the same name?

Company names are unique across the whole system, so the second company called "Nordic Logistics" to sign up is refused. This may be deliberate. It is the same shape as the Worker ID rule we just fixed, so answer it on purpose rather than by accident.

DEC-5

Self-signup: build it or delete the page?

There is a "Create an Account" page with a button that does nothing, and nothing in the product links to it. Given that email sign-in and referral registration both shipped recently, the real question is whether self-signup is on the roadmap at all.

Kept aside — features, not bugs

Real gaps, but they are new functionality rather than something to repair. They belong on the roadmap, not in this list.

FEATURE

An assessment you can save and come back to

Today a RULA, REBA or RAMP form is all-or-nothing: fill in half of it, click away or close the tab, and everything since the last save is gone with no warning. There is no autosave and no draft.

The answer is a proper in-progress state the user controls — they save what they have, the assessment shows as unfinished in the list, and they open it later and carry on from where they stopped. That is a feature to design, not a bug to patch, so it is out of this document until you schedule it.

Parked on purpose

Real findings, deliberately not scheduled. Listed so nobody re-reports them.

- **Location stops being recorded after the first session** — with location tracking on, it works for the first session after the app opens and never again. Every session after that records nothing and nothing on screen indicates it, so the data simply is not there when someone looks for it later. You chose to hold this.
- **TNO analysis drops a whole middle range of angles** — for the trunk that is 0° to 20°, most normal working posture. Farhad: skip TNO for now. If TNO is ever sold again, check this first.
- **Phone warnings versus report thresholds for the head** — Farhad wants a table comparing the two sets of criteria before deciding.
- **Three different colouring rules for one session** — pie charts, risk gauges and the measurement screen each colour the same arm differently. Farhad wants an example report and a discussion with Liyun and Micke.
- **Overlapping wording in the printed risk criteria** — optional rewording only.
- **Four invitation-flow flaws** — the admin invite path emails a dead link, a failed email locks the invite for seven days with no resend, member limits are stored but never enforced, and deleting a pending invitee can delete the person entirely. You asked to hold these.
- **Five sensor-configuration problems** — the one that matters is a recording being processed at the wrong sample rate when the phone does not report which it used, producing wrong durations and angle statistics with nothing flagging it. You asked for no tickets.

Kept separately — not bugs

DOCUMENT	WHAT IT IS
Mobile App — Improvement Tasks	The five-phase redesign brief: terminology, setup wizard, measurement screen, completion, settings. Still current.
Client Panel — v1 Restructure Tasks	The 34 tasks that bring the client panel in line with the agreed prototype. Partly delivered; still the spec.
Planned Functional Changes	The two-week customer-facing plan. Referral registration and the time-series default shipped; the demo organisation, session cleanup and the mobile items did not.
Firebase Migration Proposal · DNS Rebuild Plan	Proposals and infrastructure planning, separate tracks.
Calculation Issues — CEO Review	Farhad's written rulings. Kept as the decision record — it is the evidence behind DEC-1.

Already fixed — do not re-raise

Fifty-four items from the July and August audits, verified done in the code today. Backend items are done on `dev`, which is twenty commits ahead of `main` — so they are not in production yet. The web and mobile fixes are live.

BACKEND

A full test suite from nothing, plus sixteen findings.

Testing — the entire four-phase epic was delivered: infrastructure and fixtures, calculation logic, company separation and integration, signal processing and report maths. The backend went from 17 real test methods to a suite the pipeline runs.

Correctness — static time was 60× too small; the arm filter never ran because of a spelling mistake; decimals were chopped so "more than 30" meant "31 or more"; the Excel export wrote head values over trunk values; a trunk data situation quietly produced incomplete assessments. All fixed.

Behaviour — a generated RAMP now has its own status instead of pretending to be complete; notifications name the measurement rather than the assessment; the three notification settings that did nothing were removed; failed-measurement alerts are filed under the right company; the auto-archive rule matches the renamed statuses; merging a condition no longer fails every time; the per-minute job no longer crashes on a deleted worker; session merge was rebuilt to recompute from the merged file and delete sources only on success.

Access — support tickets and the AI report are now scoped to the caller's company; Worker IDs are unique per company with an availability check; the statistics export checks ownership. **Other** — invitations no longer share one expiry, plaintext passwords are no longer emailed, the N+1 queries on user and organisation lists are gone, the dead Slack notifier was removed, temperature prints with °C.

WEB

More than the July list suggested — the v2 work closed a lot of it.

"Measured this month" reads the real number; the shared table has a proper error state with a retry; the auth footer links point at the right pages; temperature prints with °C in both PDFs; full time series open by default; session merge is pre-validated before it is offered.

And, found during this re-check: exposure limit colours now compare real values against real thresholds and the head has its own limit; Home has loading and error states instead of showing a dashboard of zeros; the Home inbox was rebuilt so archived items are excluded and "needs an assessment" is answered by the server; the generated-RAMP status is handled properly on both sides; multi-task MEC rows are labelled with their task; the placeholder RULA dialog is gone; the notifications dropdown has its New and Earlier grouping.

MOBILE

The client's incident is closed end to end, and all four calculation corrections shipped.

The app shows the message the backend actually sent instead of "Bad request"; the Worker ID is checked before the session starts rather than after the recording; syncing adopts an existing worker instead of creating a duplicate.

All four corrections Farhad asked for are in: head bending warns at 30° instead of 45°, backward bending counts from −5° instead of any lean at all, arm speed warns at 30°/s instead of 60, and the PDF's recommended arm-speed limit reads 30°/s instead of 20. Session statuses use the values the backend expects. Calibration videos no longer exhaust the decoder, firmware update is reachable again, and any Movesense sensor is accepted.

How this was checked

- All four repositories were pulled first and read on `dev`: backend `ce515fb`, web apps `18dcf7af`, mobile `7a808f6a`, landing page `bb99528`.
- Nine documents were merged: the QA bug report (13 Jul), the general audit (20 Jul), the calculation issues (27 Jul) and Farhad's review (3 Aug), the silent-failure audit (6 Aug), the web improvement tasks (27 Jul), the backend testing epic (5 Aug), the April backlog, and the planned functional changes.
- **Every item was re-read in the current code.** An earlier draft carried some findings over from the older audits without re-checking them; eleven of those turned out to be already fixed and have been moved to the back. Two were re-graded rather than removed: the AI setting is reachable (it just has a confusing order), and the Home counts problem is much smaller than reported.
- **Removed in this revision.** "Changing rows per page breaks the list" was dropped — the control it described only exists in the old tables, and since the v2 interface was turned on everywhere in July none of those screens render any more. No screen in the product lets you change how many rows you see; adding one would be a new feature, not a fix. The orphan work-cycle steps item was downgraded to a guard after production showed zero of them.
- **Checked against production.** No worker account has ever signed in (five accounts, zero logins), twenty-three measurements belong to worker profiles, and there are no orphan work-cycle steps. Everything else was read in code only — nothing was reproduced against a running system, so the number of records affected by A2 is still unknown. The landing-page backlog was not re-audited; it is a static marketing site and its items are unchanged since April.
